

AI RESUME SCREENING SYSTEM



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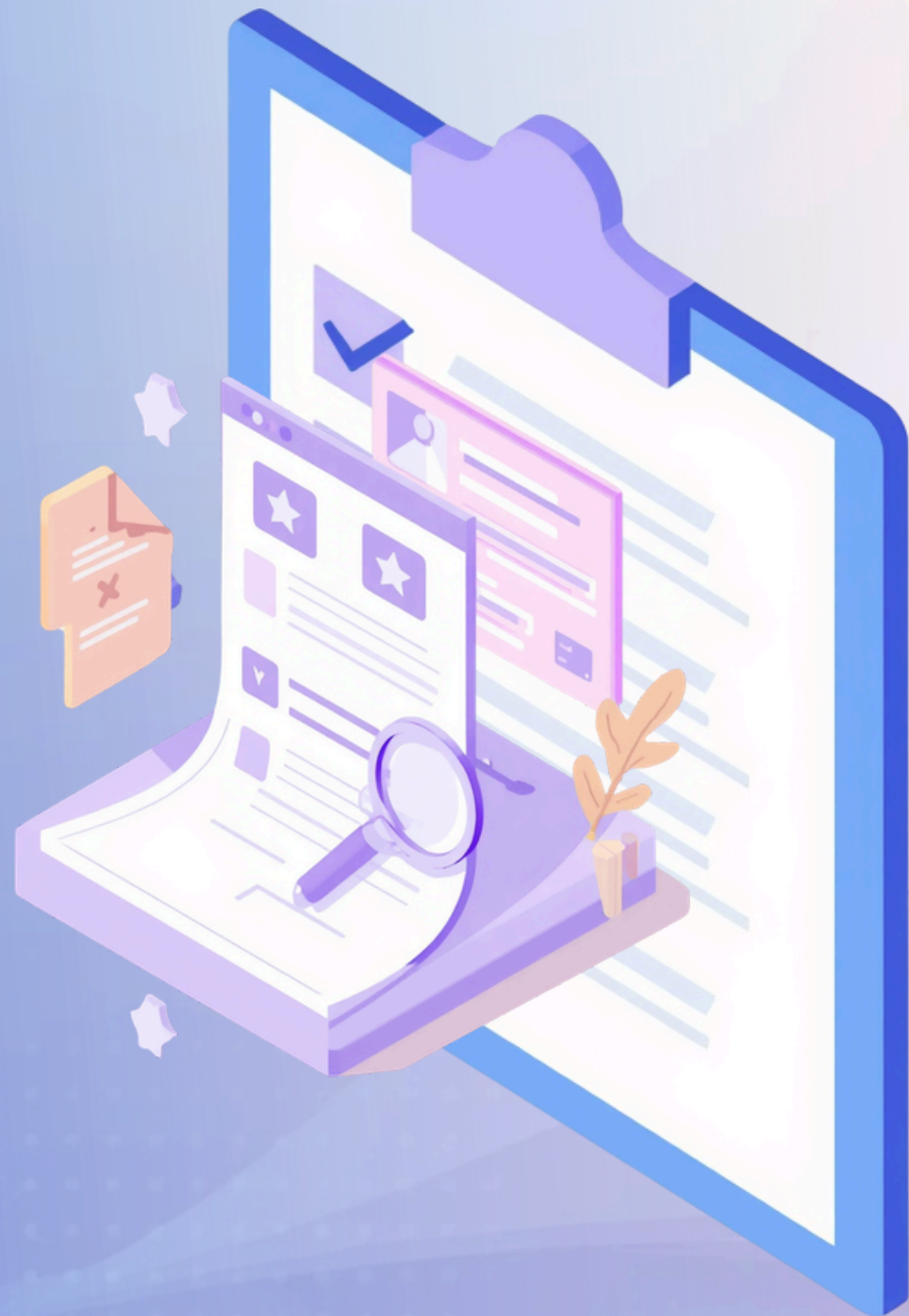


INTRODUCTION & PROBLEM STATEMENT

Every day, recruiters face hundreds to thousands of resumes for a single job opening. Manual screening is time-consuming, error-prone, and inconsistent – making it nearly impossible to identify the best candidates efficiently.

- High Volume: Organizations receive massive numbers of applications, overwhelming HR teams
- Inefficiency: Manual review leads to slow hiring cycles and missed opportunities
- Overlooked Talent: Qualified candidates are often missed due to human fatigue and bias
- Inconsistent Evaluation: Different reviewers apply different criteria, leading to unfair assessments
- Business Impact: Delayed hiring affects team productivity and organizational growth

There is a clear need for an intelligent, automated solution that can screen resumes accurately, consistently, and at scale.



PROJECT OBJECTIVES

Automate Resume Screening

Extract & Analyze Skills

Calculate Compatibility Scores

Store & Report Candidate Data



TECHNOLOGY STACK

Python & Streamlit

PyPDF2 & Pandas

Scikit-Learn & spaCy

Plotly & CSV

Python serves as the core programming language, while Streamlit powers the interactive web interface. Scikit-Learn enables TF-IDF vectorization and cosine similarity calculations for resume matching. spaCy handles NLP-based skill extraction from resumes and job descriptions. PyPDF2 extracts raw text from uploaded PDF resumes, and Pandas manages candidate data and report generation. Plotly delivers interactive visualizations and analytics dashboards, while CSV files provide lightweight candidate data storage and export functionality.



SYSTEM ARCHITECTURE

Resume Upload → Job Description Input → Text Extraction (PyPDF2) → Skill Extraction (spaCy NLP) → Text Vectorization (TF-IDF) → Cosine Similarity Calculation → Compatibility Scoring → Candidate Ranking → Data Storage (CSV/Pandas) → Report Generation (Plotly)

Each module processes data sequentially, passing structured output to the next stage. The pipeline ensures automated, consistent, and scalable resume evaluation from raw input to final ranked results and exportable reports.

PROJECT WORKFLOW

Upload Resume & Enter JD

Extract Skills & Vectorize

Score, Store & Report



A 10-step automated pipeline: from resume upload to final ranked reports – powered by NLP and Machine Learning.

MACHINE LEARNING & NLP CONCEPTS

TF-IDF (Term Frequency-Inverse Document Frequency)

Converts raw resume and job description text into weighted numerical vectors. Terms that appear frequently in a document but rarely across all documents receive higher weights, highlighting the most relevant and distinguishing skills.

Key Steps:

- Tokenize resume and JD text
- Compute term frequency per document
- Apply inverse document frequency weighting
- Output: Sparse numerical feature vectors

Result: Each resume becomes a vector of skill importance scores, ready for similarity comparison.

Cosine Similarity

Measures the angle between two vectors (resume vs. job description) in high-dimensional space. A smaller angle means higher similarity – the resume closely matches the job requirements.

Key Insight:

- Score ranges from 0 (no match) to 1 (perfect match)
- Independent of document length
- Used to rank candidates by compatibility

Result: Each candidate receives a match percentage score, enabling automated ranking and shortlisting.





KEY FEATURES

Resume Upload (PDF)

Seamlessly upload and parse PDF resumes for instant text extraction.

Job Description Input

Enter JD text to define required skills and role criteria.

Skill Extraction & Matching

NLP-powered skill detection with matched and missing skill highlights.

Compatibility Scoring & Ranking

Cosine similarity scores rank candidates by best fit automatically.

Candidate History & Leaderboard

Track all screened candidates with a ranked leaderboard view.

CSV Export & PDF Reports

Download structured candidate data and detailed PDF screening reports.



RESULTS & CONCLUSION

The AI Resume Screening System successfully automates the hiring pipeline using Machine Learning and NLP. Recruiters benefit from faster, consistent, and data-driven candidate evaluations. The system accurately extracts skills, computes compatibility scores, and generates actionable reports – reducing manual effort and improving talent discovery. Future enhancements include advanced NLP models, multi-format support, and interview scheduling automation.

Automated Screening

Accurate Skill Matching

Future: Advanced NLP



THANK
YOU.